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time it takes for displays and other such components to become available for industrial products. Also, the product cycles of typical T&M product designs are much longer than for consumer devices.

Despite challenges related to scale and T&M product longevity and design cycles, a growing number of industrial components suppliers are working to close these gaps. The use of shared product platforms can help raise volumes to interesting levels for individual vendors and through value-added suppliers of such assemblies.

On the signal conditioning front, ongoing advances in semiconductor devices and analogue and RF ICs are helping to push sensitivity, signal range and speeds when used in instrumentation front-ends. Refreshing product designs frequently and, in some cases, using modular design approaches can help enable fresh access to the latest analogue components, as well as displays and user interface technologies.

Processing power for embedded designs also continues to evolve exponentially with specialised DSP

and FPGA devices that are constantly increasing in speed. Multi-core processors are now available in industrial and mil-grade versions to allow more sophisticated user interfaces and digital signal processing to be used in medium- and lower-cost instruments.

Doing more in software rather than hardware helps reduce costs and improve functionality and flexibility for users and can also facilitate platform reuse and leverage.

Key trends driving next-generation devices

The global race to achieve useful, intuitive, elegantly integrated products is well underway in the T&M sphere, just as it is in the mobile phone and computing worlds.

Many of the advances in the next generation of instruments will be related to usability and connectivity to support the Internet-friendly generation of users. Though signal performance in T&M will always remain a top priority, human-to-machine and machine-to-machine interfaces will get some much needed attention.

To understand the implications of these interfaces for T&M products and vendors, one needs to just look at the smartphone revolution to see how product innovation and brand leadership in such devices can shift the dynamics of a very large consumer industry in just a few years.

To meet the reduced profit margins due to increased competition, T&M manufacturers will need to innovate and use new technologies of smart mobile devices and cloud computing.

While tablets and smartphones cannot replace ubiquitous PC or PC-based measurement platforms like PXI, they offer unique benefits when used as extensions to a test system. When The Nielsen Company surveyed consumers in 2011 to understand why they were using tablets instead of traditional PCs, the top reasons cited included user experience improvements like superior portability, ease of use, faster startup time and longer battery longevity. Given this information, the expected use cases for mobile devices within automated test include test system monitoring and control and test data and report viewing.

While today's technology offers solutions for monitoring or remote reporting via mobile devices, test organisations will need new expertise to unite the networking, Web services and mobile app portions of the solution.

"Development of next-generation single-chip T&M instruments using the latest extreme ultraviolet technology will help increase the operating margins without any compromise on the functionality. With the rapid rise in communication technologies like 3G and 4G, the implementation of remote testing will reduce the costs significantly," says Paras Dagli, executive director, Dynamag (India).

In short, the Internet and the seemingly never-ending demand for more mobile computing devices and capability means that future generations of T&M equipment will also continually be challenged to deliver faster, better and easier to obtain results! ●

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